

Bharath Kukka

AI & Robotics Engineer passionate about creating autonomous systems and intelligent applications, Skilled in machine learning, robotics, and computer vision to deliver real-world solutions.

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Internship

Robotics Engineering Intern | 6-Axis Manipulator 

Manav Rachna University

Jun 2025 - Present

- Lead a 3-member team in the end-to-end design, development, and integration of a 6-axis robotic manipulator.
- Model and validate the manipulator’s complex kinematics and dynamics using the MATLAB Robotics Toolbox.
- Engineer the core software architecture using ROS2 and develop path planning and trajectory planning algorithms.
- Implementing CV/ML models for real-time object detection on edge hardware.

Projects

ShunyaNet (Multi-Domain Custom CNN Architecture) 

Apr 2025 - Present

- Architected and developed “ShunyaNet,” a custom, hybrid CNN by researching and integrating state-of-the-art (SOTA) modules, including InceptionBlocks, SEBlock, and Self-Attention.
- Built an 8-class emotion-recognition model in PyTorch to drive a supportive content recommendation engine.
- Ported and trained TensorFlow/Keras versions for crop disease detection (Cotton 4-class and Paddy 10-class).

Home Security System 

Jan 2025 - May 2025

- Built a 3-class face-recognition access-control system using MTCNN and InceptionResnetV1, with a threat-detection module for unknown individuals that triggers weapon alerts via Gemini API and Telegram.
- Currently developing an in-home fall-detection model to monitor and trigger high-priority alerts to family and emergency services.
- Working on an end-to-end security platform that combines three CV models with a real-time web dashboard for monitoring and user control.

4-Axis Manipulator 

Jul 2024 - Dec 2024

- Led a 3-member team in the end-to-end design, build, and implementation of a 4-axis robotic manipulator.
- Modelled and validated the robot's kinematics by deriving DH parameters and solving forward/inverse kinematics using the MATLAB Robotics Toolbox.
- Developed a Python (Tkinter) GUI to control the arm, sending serial commands to an Arduino Uno, managing stepper motors, and servos.

Skills

Languages: Java, Python
ML/CV: TensorFlow, Keras, PyTorch, OpenCV, scikit-learn
Developer Tools: PyCharm, Eclipse, MATLAB, Docker
Robotics: ROS2, MATLAB Robotics Toolbox
Version Control: Git, GitHub

Education

B.Tech in Robotics & AI	Manav Rachna University	CGPA: 8.6	Jun 2022 - Jun 2026
Senior Secondary	Sri Chaitanya Junior College	Percentage: 94.4%	Jun 2020 - May 2022

Research

- Co-authored and presented “Design & Implementation of 5DOF Robotic Arm” at ICIACS25.



May 2025

Certifications

- Python Web Developer | Infosys Foundation - **Aug 2025**
- Robotics & Control | NPTEL - **Mar 2024**
- Java | Coding Ninjas - **Jun 2025**

Achievements

- Dean’s List in 4th and 6th Semesters.

Languages : Telugu, Hindi, English.